

Multi-omic relay velocity modeling uncovers dynamic chromatin-transcription regulation across cell states

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Cell fate transitions emerge from dynamic gene expression programs, yet existing RNA velocity models primarily rely on RNA abundance and globally inferred latent time, limiting their ability to capture local regulatory dynamics. To address these limitations, we introduce MoFlow, a deep neural network that integrates multi-omic data within a relay velocity framework. Unlike previous approaches, MoFlow flexibly infers velocity parameters at single-cell resolution without pre-assigned latent time, enabling a comprehensive and locally adaptive estimation of gene expression kinetics. Applied to single cell multi-omic datasets from brain, skin, and blood cells, MoFlow distinguished chromatin-dependent and independent transcriptional regulation, validated transcription repression models, and identified asynchronous gene repression in neural progenitors. It also uncovered transcriptional activation of DNA damage response genes in radial glia with distinct subnuclear localization. By resolving fine-grained regulatory programs, MoFlow advances the interpretability and precision of RNA velocity analysis beyond the limits of existing models.

Cell fate transitions are dynamically regulated by transcriptional programs and epigenetic modifications^{1–3}. Single-cell RNA sequencing (scRNA-seq) has enabled unprecedented resolution of cellular heterogeneity, allowing the identification of subpopulations and dissection of transcriptional programs at the single-cell level^{4,5}. However, scRNA-seq provides only static snapshots of gene expression, limiting our ability to infer the dynamic trajectories underlying these transitions. RNA velocity addresses this limitation by estimating each cell's future transcriptional state based on the ratio of unspliced to spliced RNA transcripts⁶.

Traditional RNA velocity inference methods fall into two main types: ordinary differential equation (ODE)-based models and neural network-based approaches. ODE models infer gene-specific kinetic rates and latent time^{6,7}, but often assume a single transcriptional path per gene and perform poorly in heterogeneous systems. Neural

models like VeloAE⁸, VeloVAE⁹, and VeloVI¹⁰ learn latent spaces that capture complex structures, but they typically impose a global temporal ordering and assume synchronous transcription and splicing. Both approaches rely only on RNA abundance and do not incorporate regulatory information such as chromatin accessibility, which limits their ability to model gene regulation in context.

Recent extensions seek to overcome some of these limitations. cellDancer¹¹ introduces a local relay velocity model without latent time, but is limited to transcriptomic data. MultiVelo¹² incorporates chromatin accessibility in an ODE-based model but assumes fixed gene classifications and shared kinetic regimes. Despite these advances, three key challenges remain unresolved: (i) modeling dynamics with branching trajectories without fixed gene labels, (ii) capturing asynchronous transcriptional kinetics, and (iii) integrating epigenomic context without relying on rigid structural assumptions.

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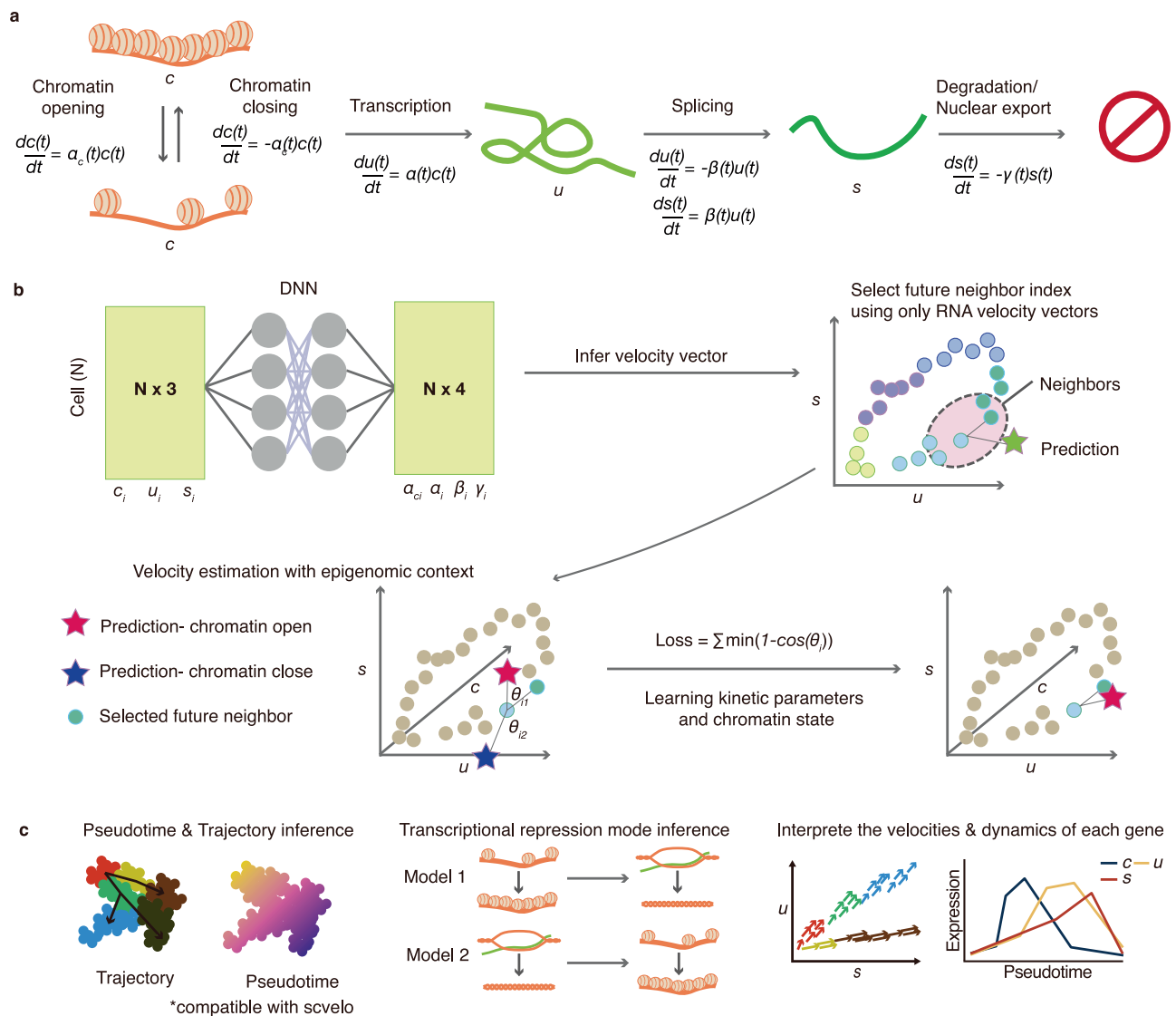


Fig. 1 | Overview of MoFlow. a Conceptual model of gene regulatory dynamics incorporating chromatin accessibility. The framework models chromatin opening and closing as time-dependent processes, alongside transcription, splicing, and degradation of RNA. **b** Architecture of the MoFlow deep neural network. For each gene, chromatin accessibility (c_i), unspliced (u_i), and spliced (s_i) RNA levels for each cell i are used to infer cell-specific kinetic parameters. Kinetic rates ($\alpha_{ci}, \alpha_i, \beta_i, \gamma_i$) are learned in a cell-specific manner, allowing MoFlow to capture heterogeneous and asynchronous gene regulation. Future neighbors are selected using RNA velocity

vectors, and velocity vectors under both chromatin opening and closing assumptions are compared using cosine similarity. The lower-loss scenario is used to guide learning, enabling flexible and context-aware estimation of gene expression dynamics. **c** Downstream analysis of MoFlow. Velocities can be projected and visualized, and velocity pseudotime can be computed using the scVelo package. In addition, the framework allows detailed examination of individual gene dynamics and transcriptional repression modes.

To address these gaps, we introduce MoFlow, a deep neural framework that combines a locally adaptive relay velocity model with multi-omic integration. MoFlow leverages chromatin accessibility and transcriptomic profiles to infer cell-specific kinetic parameters—including rates of chromatin opening and closing, transcription, splicing, and degradation—without relying on predefined latent time or fixed gene classifications. At its core, MoFlow aligns each cell to its most probable future neighbors based on local transcriptomic and epigenomic context, thereby allowing for asynchronous kinetics modeling accommodating branch-trajectories. This relay-based alignment replaces global time inference, enabling flexible modeling of heterogeneous developmental processes while explicitly incorporates epigenomic regulation.

We applied MoFlow to multi-omic datasets from the developing human brain, embryonic mouse brain, mouse skin, and human blood cells. MoFlow captured lineage-specific transcriptional dynamics, such

as asynchronous repression and chromatin-independent transcriptional bursts. It also identified diverse RNA velocity states, such as decoupling between transcription and splicing, and revealed temporal lags between chromatin remodeling and transcriptional activation consistent with RNA export kinetics. Compared to existing methods, MoFlow produced more biologically coherent trajectories, improved directional accuracy, and more reliably resolved cell-type-specific regulatory programs across diverse developmental systems.

Results

MoFlow: deep learning framework to estimate cell-specific RNA kinetics incorporating chromatin accessibility

MoFlow is a deep learning framework designed to estimate cell-specific transcriptional kinetics by integrating chromatin accessibility with RNA velocity. As illustrated in Fig. 1a, the model extends traditional RNA velocity by introducing four kinetic parameters—chromatin

accessibility change (α_c), transcription rate (α), splicing rate (β), and degradation rate (γ)—all inferred at single-cell resolution. We describe the system dynamics using the following equations:

$$\frac{dc}{dt} = \alpha_c \cdot (k - c) \quad (1)$$

$$\frac{du}{dt} = \alpha \cdot c - \beta \cdot u \quad (2)$$

$$\frac{ds}{dt} = \beta \cdot u - \gamma \cdot s \quad (3)$$

Here, k is a binary variable representing chromatin state (1 for opening, 0 for closing), and c , u , and s denote chromatin accessibility, unspliced RNA, and spliced RNA, respectively. As shown in Fig. 1b, a deep neural network is used to infer these parameters from per-cell chromatin and RNA measurements (c_i , u_i , and s_i), enabling kinetic modeling at single-cell resolution. The model employs a locally adaptive relay strategy to align each cell with its most probable future neighbor based on RNA velocity vectors, replacing global latent time with flexible, context-aware alignment.

Inspired by the relay-based design of cellDancer¹¹, MoFlow minimizes the cosine distance between the predicted velocity vector and observed transcriptomic progression to guide parameter learning. To handle regulatory ambiguity, the model evaluates predictions under both chromatin opening and closing assumptions, selecting the direction with lower angular error. This comparative loss formulation enables chromatin-aware modeling without relying on fixed gene classifications or globally shared kinetic regimes. To account for less dynamics of chromatin accessibility relative to RNA, MoFlow adopts a two-stage training procedure: the model first learns RNA-based kinetic parameters with neighborhood alignments, followed by joint optimization of chromatin-related parameters while holding those alignments constant. Full training procedures, including loss formulations and architectural details, are provided in “Method” 5.

To ensure compatibility with existing Python-based single-cell and velocity analysis toolkits, the output of MoFlow is provided in the AnnData format (Fig. 1c). This enables seamless use of scVelo for velocity embedding and flow visualization, as well as for the computation of velocity pseudotime. Beyond global trends, the framework also supports gene-level analyses, enabling detailed inspection of transcriptional dynamics for individual loci.

MoFlow reveals continuity between two distinct transcriptional repression mechanisms

We first applied MoFlow to 10x Multiome data from the developing human brain cortex¹³. By integrating chromatin accessibility with gene abundance, MoFlow inferred cell-specific velocity vectors of 842 genes across 4693 cells, enabling the inference of a velocity stream that closely aligns with the well-characterized trajectory of human brain development (Fig. 2a). MoFlow accurately reconstructed the developmental trajectory, assigning early pseudotime values to proliferative Cyc. Prog. cells (Fig. 2b). Notably, the G2/M scores show a clear decline along pseudotime, supporting the interpretation that MoFlow captures the transition from proliferative progenitors toward differentiated neuronal states (Fig. 2c)^{7,14}. Furthermore, MoFlow achieved high cross-boundary direction correctness (CBDir) scores (MoFlow: 0.362; MultiVelo¹²: 0.211; cellDancer¹¹: −0.015; scVelo⁷: 0.211), suggesting that it reliably captured directionality of transitions across transcriptional boundaries (Fig. 2d and Supplementary Table 1).

Unlike scVelo or cellDancer, which erroneously inferred a backflow trajectory from neuronal induced IPC/GluN1 neurons (red) to Cyc. Prog. (blue) or RG (orange), MoFlow avoided inferring these biologically implausible transitions (Fig. 2a). Moreover, MultiVelo tended to

impose overly linear dependencies among differentiating populations, resulting in a misleading trajectory from GluN3 (brown) to GluN2 (purple; upper-layer like cluster) and further to GluN4/5 (pink/gray; deeper-layer like cluster). This is inconsistent with established neurodevelopmental patterns, where upper-layer neurons develop after deeper-layer neurons, and direct transitions between them are not observed^{15,16}. In contrast, MoFlow correctly inferred parallel and independent differentiation paths for GluN2 and GluN4/5, in agreement with known laminar development.

Previous models typically predefine gene-specific transcriptional trends based on the observed distribution of u and s , and estimate kinetic parameters under those fixed assumptions. In contrast, MoFlow dynamically adjusts these trends by minimizing the discrepancy between predicted velocities and local neighborhood transitions during training. These results align more closely with the actual differentiation patterns of the cells (Fig. 2e for MoFlow, MultiVelo and cellDancer; Supplementary Fig. 1a, b including scVelo). For example, *MKI67* is a marker for cellular proliferation, which is maximally expressed in G2 phase and mitosis¹⁷. Its expression is highest in Cyc. Prog. and should decrease as cells progress toward RG. MultiVelo and scVelo, however, predict the opposite pattern—an increase from RG to Cyc. Prog.—indicating that it fails to capture the expected trend. *SDK2*, which encodes a synaptic-adhesion molecule, should be up-regulated as cells differentiate toward glutamatergic neurons; cellDancer, however, predicts the opposite, with expression falsely declining along this trajectory¹⁸. Likewise, *CREB5*—normally elevated in mGPC/OPC—is expected to increase as cells progress from RG to mGPC/OPC, yet cellDancer instead models its expression as decreasing toward RG¹⁹.

Li et al. described two kinetic scenarios for transcriptional repression¹². Model 1 posits that chromatin closure causes transcriptional silencing, whereas model 2 permits transcriptional silencing to initiate prior to chromatin compaction. This model infers gene-specific switching times and latent times for each gene by determining which sequence–chromatin closing followed by RNA repression, or vice versa—best fits the data.

Unlike the ODE-based framework, MoFlow does not explicitly infer switching or latent times. Instead, for each gene, we evaluate the likelihood that its observed chromatin and transcription dynamics following each model: the m_1 score, which reflects increasing RNA despite chromatin closing during transcription repression, and the m_2 score, which reflects decreasing RNA despite chromatin opening during transcription repression (see “Methods” for formal definitions). As expected, genes that MultiVelo assigns to model 1 show significantly higher m_1 scores in MoFlow, whereas model 2 genes exhibit higher m_2 scores (Fig. 2f, g). MultiVelo’s discrete state labels (“on”, “off”, “complete”) also align with MoFlow’s continuous RNA-on and RNA-off fractions, which respectively denote the proportion of cells showing coordinated activation or repression of RNA velocities (Fig. 2h, i), indicating that MoFlow reproduces the same regulatory logic without relying on explicit switching times or latent time during estimation.

Notably, genes assigned to model 2 in MultiVelo exhibited significantly higher RNA-off score, the fraction of cells with negative velocity for both unspliced and spliced RNA (Fig. 2j). Interestingly, MultiVelo previously reported that model 2 genes are enriched for mitosis-related functions. This observation is consistent with our finding that such genes are predominantly expressed in Cyc. Prog. cells and progressively downregulated as cells transition toward RG states. Given that active cell division is largely restricted to progenitor populations and that transcriptional repression of mitotic genes precedes chromatin closure in model 2, the observed dynamics align well with the biological trajectory of early neurodevelopment. Together, these results demonstrate that MoFlow not only reproduces known regulatory logic without relying on explicit switching models, but also

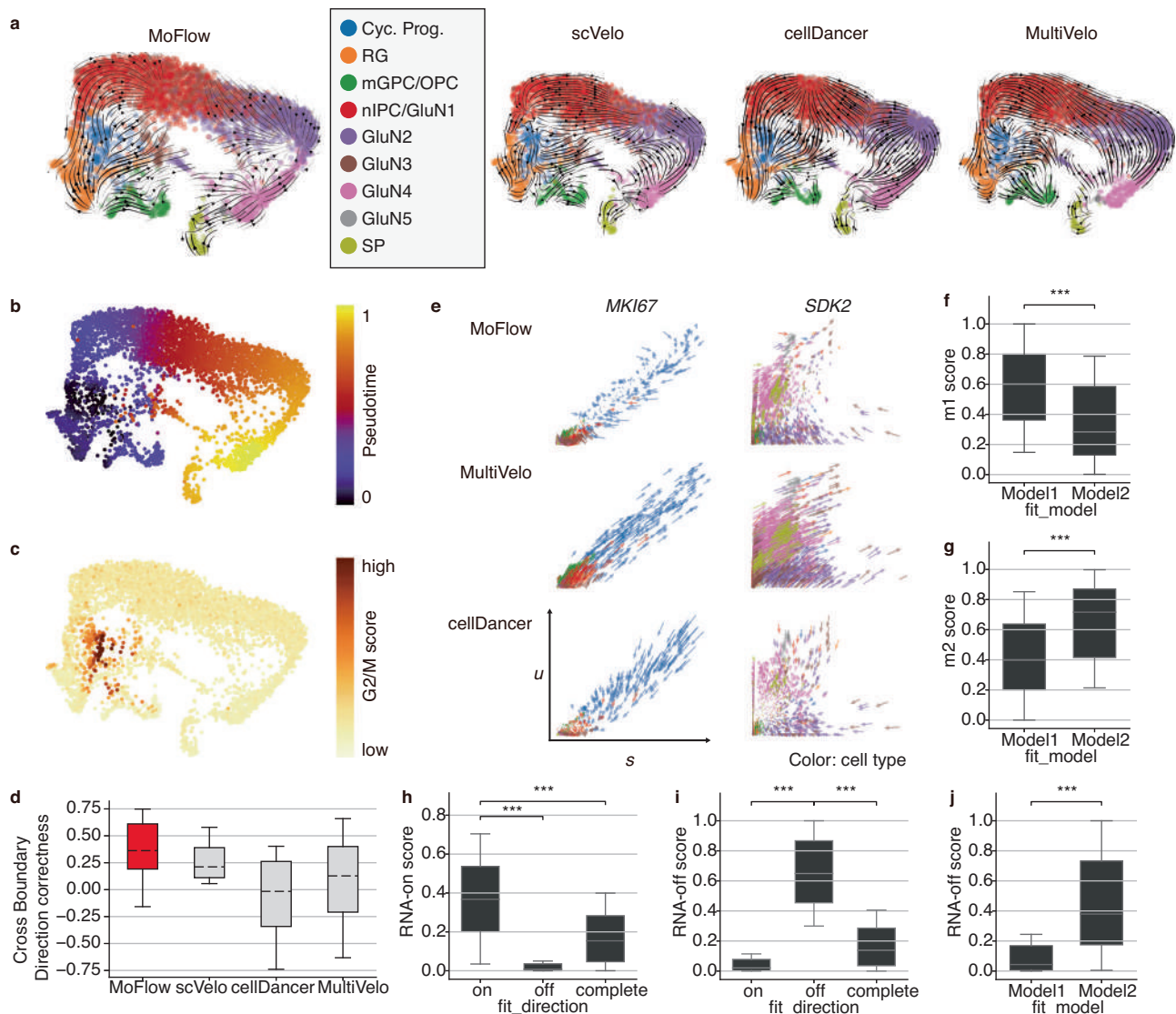


Fig. 2 | MoFlow reveals accurate developmental trajectories and regulatory dynamics in the human brain cortex. **a** RNA velocity streamlines inferred by MoFlow, bluescVelo, cellDancer, and MultiVelo on the same UMAP embedding, colored by cell type. MoFlow reconstructs trajectories consistent with known developmental hierarchies, while others exhibit biologically implausible transitions. **b** Pseudotime inferred by MoFlow reflects a coherent developmental gradient. **c** G2/M score distribution mapped onto the UMAP embedding reveals elevated proliferative activity in Cyc. Prog. cells (blue color at **a**), which aligns with pseudotime and supports the biological validity of the inferred transitions. **d** MoFlow achieves the highest cross-boundary direction correctness (CDBir), indicating improved inference of lineage progression. The boundaries are defined by the first and third quartiles, with the whiskers extending to 1.5 times the interquartile range. The mean is marked at the center of the plot. **e** Velocity patterns of *MKI67*, *SDK2* across models. MoFlow correctly captures expected dynamics based on known marker gene expression. **f**, **g** Distribution of m_1 scores

(reflecting increasing RNA despite chromatin closing during transcription repression) and m_2 scores (reflecting decreasing RNA despite chromatin opening during transcription repression) derived from MoFlow, which align with MultiVelo's model 1 and model 2 gene classifications, respectively. Two sided t-test was performed, and boundaries are defined by the first and third with whiskers extending 0.5 times the interquartile range. The median is marked at the center. **h**, **i** Distribution of RNA-on scores (fraction of cells with coordinated RNA increasing) and RNA-off scores (fraction of cells with coordinated RNA decreasing) across genes stratified by transcriptional state category ("on", "off", "complete") assigned by MultiVelo. Two sided t-test was performed, and boundaries are defined by the first and third, with whiskers extending 0.5 times the interquartile range. The median is marked at the center. **j** RNA-off scores further support transcriptional shutdown in model 2 genes, consistent with chromatin-first or RNA-first repression regimes. *** denotes $p < 0.001$.

provides finer-grained, biologically grounded interpretations of transcriptional repression during early neurodevelopment.

MoFlow captures transcriptional decoupling and asynchronous regulation in human OPC lineages

To examine the biological plausibility of MoFlow's pseudotime prediction, we categorized genes into seven clusters based on their transcriptional changes along the pseudotime trajectory (Fig. 3a). Functional annotation analysis using DAVID²⁰ based on biological

process (BP) ontology within the Gene Ontology Resource^{21,22} revealed a sequential cascade of neurodevelopment. Early clusters (0–1) were enriched for terms related to cell cycle and lineage priming, such as cell division ($p = 1.63e^{-15}$) or mitotic cell cycle ($p = 2.46e^{-6}$). Intermediate clusters (2–3) reflected proliferation and neuron projection development pathways (e.g., positive regulation of cell migration, $p = 0.038$, actin filament organization, $p = 0.008$), while late clusters (4–6) featured signatures of synapse formation, excitatory signaling, and neuronal differentiation (e.g., nervous

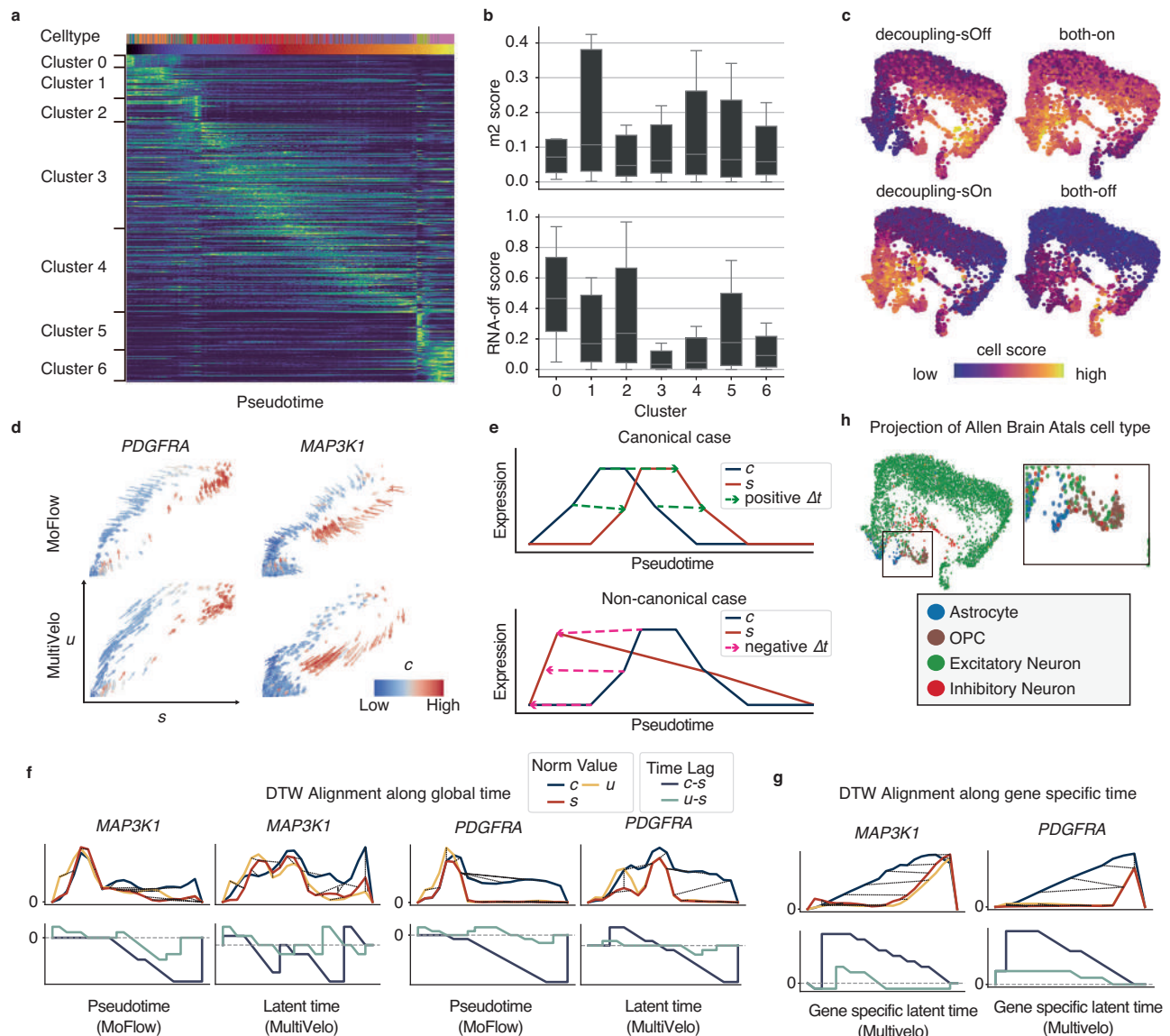


Fig. 3 | MoFlow reveals transcriptional decoupling and asynchronous regulatory dynamics in OPC lineage. **a** Genes were clustered based on transcriptional velocity patterns along MoFlow-derived pseudotime. **b** Distribution of m_2 score (top) and RNA-off score (bottom) along gene clusters. The boundaries are defined by the first and third quartiles, with the whiskers extending to 0.5 times the inter-quartile range. The median is marked at the center of the plot. **c** Distribution of RNA velocity-based transcriptional states of cells. Decoupling-sOff (unspliced $v^u \geq 0$, spliced $v^s < 0$) states are enriched in mGPC/OPC regions. **d** Gene-wise velocity streamlines of *PDGFRA* and *MAP3K1* under MoFlow and MultiVelo. Both genes

exhibit decoupling-sOff dynamics in OPCs. **e** Scheme of negative or positive time-lag derived from DTW analysis. **f** DTW alignment of chromatin (c), unspliced (u), and spliced (s) signals from *PDGFRA* and *MAP3K1* under MoFlow pseudotime and MultiVelo global latent time. **g** DTW alignment under MultiVelo's gene-specific latent time. **h** Adult brain scRNA-seq cells projected onto MoFlow embedding using annotations from Allen Brain Atlas. OPC-aligned cells localize to regions with decoupling-sOff signatures, and chromatin bursts in *PDGFRA* and *MAP3K1*, supporting the biological relevance of lag-based regulatory transitions.

system development, $p = 0.049$, presynaptic modulation of chemical synaptic transmission, $p = 0.051$). These results indicate that MoFlow accurately captures the transition from proliferative progenitors to mature neurons, providing an interpretable and biologically grounded view of transcriptomic dynamics.

We then investigated the regulatory mechanisms associated with each cluster, and we computed the average m_2 and RNA-off scores (Fig. 3b and Supplementary Fig. 2a, b). Clusters 0 and 1—associated with mitosis and cell proliferation—showed markedly elevated RNA-off or m_2 scores and decreased RNA-on score. In contrast, m_1 scores remained relatively consistent across all clusters, indicating no significant cluster-specific differences characteristic of model 1-type transcriptional repression. These patterns suggest early and coordinated repression of cell cycle-related genes as cells exit

the proliferative state. Notably, these genes were also frequently classified as model 2 by MultiVelo, supporting the view that their repression occurs asynchronously, potentially independent of chromatin remodeling.

To characterize whether the states of different genes are temporally coordinated, we categorized gene dynamics within each cell based on the sign of unspliced and spliced RNA velocities (v^u , v^s). We defined four distinct RNA states of cells: both-on ($v^u \geq 0$, $v^s \geq 0$), both-off ($v^u < 0$, $v^s < 0$), decoupling-sOff ($v^u \geq 0$, $v^s < 0$), and decoupling-uOff ($v^u < 0$, $v^s \geq 0$). By counting the number of genes in each state per cell, we determined the consistent cascade of transition through states; start with decoupling-sOff to decoupling-sOn or both-on, and end with both-off (Fig. 3c). In contrast, mGPC/OPC populations showed a heterogeneous mixture of states, indicative of

asynchronous or gene-specific regulatory modes. This pattern may reflect variability in inputs, suggesting that differentiation in these populations does not proceed uniformly.

This regulatory heterogeneity is exemplified by genes, such as *PDGFRA* and *MAP3K1* (Fig. 3d). Both genes displayed sustained unspliced RNA production ($v^u \geq 0$) despite declining spliced levels ($v^s < 0$), indicative of prolonged transcription following a transient burst in chromatin accessibility. While *PDGFRA* is a well-established OPC marker²³, *MAP3K1* is involved in MAPK/JNK signaling and may play a previously underappreciated role in OPC lineage specification²⁴.

To test whether the inferred regulatory layers follow a biologically plausible temporal order, we applied dynamic time warping (DTW)^{25,26} to c , u , and s profiles for representative genes, including *PDGFRA* and *MAP3K1*, aligned along both MoFlow pseudotime and MultiVelo latent time (Fig. 3e, f; details in “Methods” 5). DTW is a nonlinear alignment method that quantifies the optimal time lag between two signals, allowing us to identify asynchronous transitions that would be missed by static or linear comparisons.

Biologically, a positive c – s lag indicates that chromatin accessibility changes precede transcriptional output, consistent with canonical gene activation (Fig. 3e). Conversely, a negative lag suggests that RNA expression changes before chromatin remodeling, potentially reflecting asynchronous repression, decoupled regulation, or delayed chromatin closure. For both genes, we observed negative c – s lags under MoFlow’s global pseudotime, suggesting that transcription may be active regardless to chromatin remodeling. Also, similar inversions also appeared under alignment with global latent time inferred by MultiVelo.

Notably, when MultiVelo’s gene-specific latent time was used instead, those negative lags largely disappeared, and the trajectories aligned with the expected $c \rightarrow u \rightarrow s$ sequence, showing strictly positive c – s lags (Fig. 3g). This contrast suggests that per-gene time fitting may over-correct trajectories to match expected biological order—even when data indicate otherwise.

To determine whether these reversals were systematic, we quantified the proportion of time bins with opposing lag signs between global and gene-specific time. Over 400 genes exhibited reversals in at least 25% of bins, and 129 genes showed consistent reversal in more than 75% of bins (Supplementary Fig. 2c, d). These results underscore a limitation of latent time models: enforcing gene-wise alignment to idealized trajectories may introduce artificial temporal orderings. In contrast, MoFlow avoids this by inferring velocity without relying on latent time, enabling recovery of asynchronous and branching regulation.

Finally, to contextualize these transcriptional lag patterns within the mGPC/OPC developmental domain, we projected adult scRNA-seq profiles onto the embedding (Fig. 3h)²⁷. Cells annotated as OPCs predominantly aligned with regions exhibiting decoupling-sOff states, consistent with widespread transcriptional shutdown following unspliced RNA accumulation (Fig. 3c). Quantitatively, among mGPC/OPC cells in the decoupling-sOff region—defined as cells where 50% of genes exhibit decoupling-sOff velocity—63% were embedded as OPCs, while only 20% were OPCs in the complementary region. These OPC-aligned areas also coincided with sharp chromatin accessibility increases in *PDGFRA* and *MAP3K1* (Supplementary Fig. 2e). In contrast, astrocytes mapped to distinct transcriptional zones lacking this decoupling signature. These results support the idea that RNA-chromatin lags reflect real biological regulation, not modeling artifacts.

Deciphering the basis of transcriptional boost genes at single-cell resolution

Next, we applied MoFlow to SHARE-seq data from mouse skin, which captures transcriptional and chromatin accessibility dynamics during hair follicle differentiation²⁸. TAC cells (red/purple) are known to

differentiate into the hair shaft cuticle/cortical layer (blue), IRS (orange), or medulla (green)²⁹. MoFlow accurately reconstructed the velocity field consistent with these known trajectories, capturing lineage-specific directionality from TAC-1 through TAC-2 toward distinct terminal states (Fig. 4a). Interestingly, G2/M and S phase scores revealed elevated cell cycle activity in TAC-1 cells (Fig. 4b, c), suggesting that MoFlow assigns root-like states to actively cycling progenitors.

The dynamics of RNA-on were generally greater than those of RNA-off (Fig. 4d), which aligns with previous findings reported in MultiVelo (Fig. 4e)¹², indicating the predominance of induction-only genes concurrent with active transcription. Additionally, the distribution of m_1 scores was generally higher than that of m_2 scores, consistent with the observation in MultiVelo that model 1 genes are more abundant than model 2 genes. However, it is noteworthy that the m_1 scores of model 1 genes were higher than those of model 2 genes, whereas the m_2 scores of model 2 genes surpassed those of model 1 genes (Fig. 4f).

Building on these results, we leveraged MoFlow’s ability to estimate gene-specific kinetic parameters at the single-cell level, enabling effective analysis of multiple-rate kinetics (MURK) genes, including transcriptional boost genes³⁰. Previous work using relay velocity models showed cell-type-specific variability in transcription rates (α) among MURK genes¹¹. MoFlow refines this by incorporating chromatin accessibility, allowing decomposition of total transcription rate ($c \cdot \alpha$) into contributions from c and α .

Although MURK genes have not been systematically studied in skin, the widespread presence of induction kinetics suggests the possibility of similar amplification mechanisms. MoFlow successfully identified transcriptional boosts, exemplified by *Heph1L*, a gene over-expressed in hair follicles³¹, and outperformed cellDancer in reconstructing its trajectory (Fig. 4g). Furthermore, pronounced variation in transcription rate ($\alpha \cdot c$) within hair shaft-cuticle cells suggests dynamic modulation beyond chromatin accessibility alone.

To investigate the basis of transcriptional boosts, we computed DAC (distribution of average across cell types) scores for α and chromatin accessibility c across all genes (see “Methods”). Gene set enrichment analysis (GSEA)³² revealed that genes with high DAC for either α or c were enriched for programs related to cell cycle and differentiation (Supplementary Fig. 3a, b).

We next classified genes into four groups based on high (H) or low (L) DAC values of chromatin accessibility (c , C) and transcription rate (α , A), using a threshold of 0.05: HCHA, HCLA, LCHA, and LCLA (Fig. 5a). Functional annotation analysis using BP ontology indicated distinct enrichment patterns across four groups^{20–22}. HCHA genes were involved in active differentiation programs such as keratinocyte fate and hair follicle development (Supplementary Fig. 3c). HCLA genes were enriched for structural and ion transport processes, suggesting chromatin priming without active transcription (Supplementary Fig. 3d). LCHA genes are prepared to rapidly adjust transcriptional regulation in response to specific environmental cues, even though chromatin structure remains relatively unchanged. Notably, these genes are associated with processes, such as chromosome segregation and spindle organization, suggesting a potential role in cell cycle-specific transcriptional regulation (Fig. 5b).

To test whether LCHA genes exhibit transcriptional flexibility prior to lineage commitment, we assessed the association between their expression and cell cycle scores. Using the Mann–Whitney U test, we found that LCHA genes had significantly stronger correlations with G2/M scores compared to other groups (Fig. 5c), suggesting that chromatin-stable but transcriptionally variable genes play roles in cell cycle-dependent transitions. This suggests that LCHA genes are poised for rapid transcriptional modulation, rather than epigenetic regulation, in response to intrinsic state changes. By contrast, LCLA genes exhibited the highest p -value, consistent with their transcriptionally

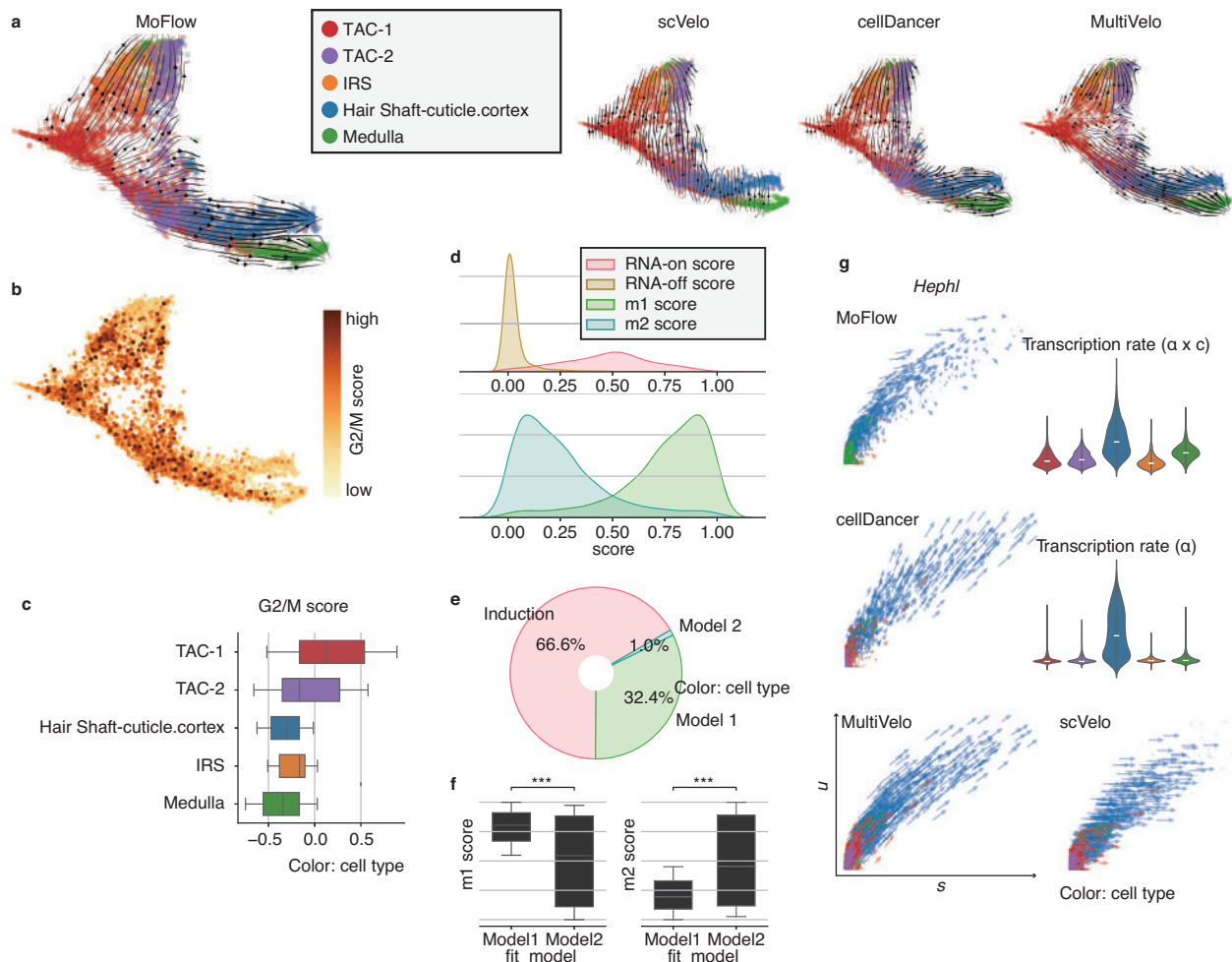


Fig. 4 | MoFlow captures transcriptional boosts and cell-cycle-linked dynamics during hair follicle differentiation. **a** Velocity streamlines inferred by MoFlow, scVelo, cellDancer, and MultiVelo on the SHARE-seq mouse skin dataset. MoFlow accurately captures lineage flows from TAC (transit amplifying cell; red/purple) toward IRS (inner root sheath; orange), medulla (green) or hair shaft-cuticle cortex (blue). **b** G2M scores mapped to the UMAP embedding reveal high proliferative activity in overall TAC-1. **c** Distribution of G2M scores across TAC and lineage clusters. TAC-1 exhibits the highest G2M scores, suggesting root-state assignment by MoFlow. The boundaries are defined by the first and third quartiles, with the whiskers extending to 1.5 times the interquartile range. The median is marked at the

center of the plot. **d** Distribution of RNA-on, RNA-off, m_1 , m_2 scores derived from MoFlow. **e** Pie charts showing proportions of genes assigned by MultiVelo: induction-only, repression-only, or combined dynamic models, consistent with model 1 and model 2 definitions. **f** Distribution of m_1 scores and m_2 scores derived from MoFlow, which align with MultiVelo's model 1 and model 2 gene classifications, respectively. Two sided t-test was performed, and boundaries are defined by first and third quartiles, with whiskers extending 1.5 times the interquartile range. The median is marked at the center. **g** The velocities derived from MoFlow, cellDancer, MultiVelo and scVelo for *Heph1L*. The violin plots showing of transcriptional activity predicted by MoFlow or cellDancer across cell types. *** denotes $p < 0.001$.

and epigenetically quiescent nature and limited involvement in active cell cycle regulation.

MoFlow estimated cell type-specific velocity parameters without requiring explicit annotations, revealing transcriptional regulation driven by distinct combinations of c and α . For example, LCHA genes, such as *Padi3* and *Myo10*, were correctly predicted by MoFlow to differentiate toward medulla and IRS, respectively, despite low chromatin accessibility in those regions (Fig. 5d, e). *Padi3* modulates keratinization via targets, such as filaggrin and trichohyalin, supporting its role in hair shaft and IRS development³³. Likewise, *Myo10*, involved in cell migration and morphogenesis, shows elevated α in TAC-2 and IRS during follicle development³⁴. In contrast, MultiVelo failed to capture these transitions due to its limited modeling of α variability (Supplementary Fig. 3e). Interestingly, although those genes are characterized to show relatively stable chromatin accessibility across cell types, cellDancer failed to reconstruct their transcriptional trajectories. It suggests that accurate modeling of RNA velocity requires not only chromatin-level modulation of transcription, but also precise inference of chromatin dynamics themselves.

Among HCHA genes, *Trps1* was successfully predicted by both MoFlow and MultiVelo, whereas *Notch1*—despite its known role in follicular fate decisions³⁵—was only accurately modeled by MoFlow. Finally, for the HCLA gene *Wnt3*, both MoFlow and MultiVelo predicted the known transcriptional increasing pattern well³⁶. In the case of cellDancer and scVelo, they failed to correctly predict the differentiation direction for *Trps1* or *Wnt3*. Together, these findings highlight MoFlow's ability to disentangle transcriptional regulation from chromatin context, outperforming models that lack dynamic or multi-omic integration.

MoFlow reveals transcriptional dynamics and subpopulation heterogeneity in radial glia

We utilized the 10x Multiome dataset of the embryonic day 18 (E18) mouse brain. In the developing mouse brain, differentiation follows a trajectory from Cyc. Prog. (blue), through RG (purple) and IPC (red) to various neuronal subtypes. In this dataset, IPCs differentiate into excitatory ExM (green) and SP (brown), further diverging into UL (pink) and DL (orange) neurons³⁷. Notably, the dataset primarily

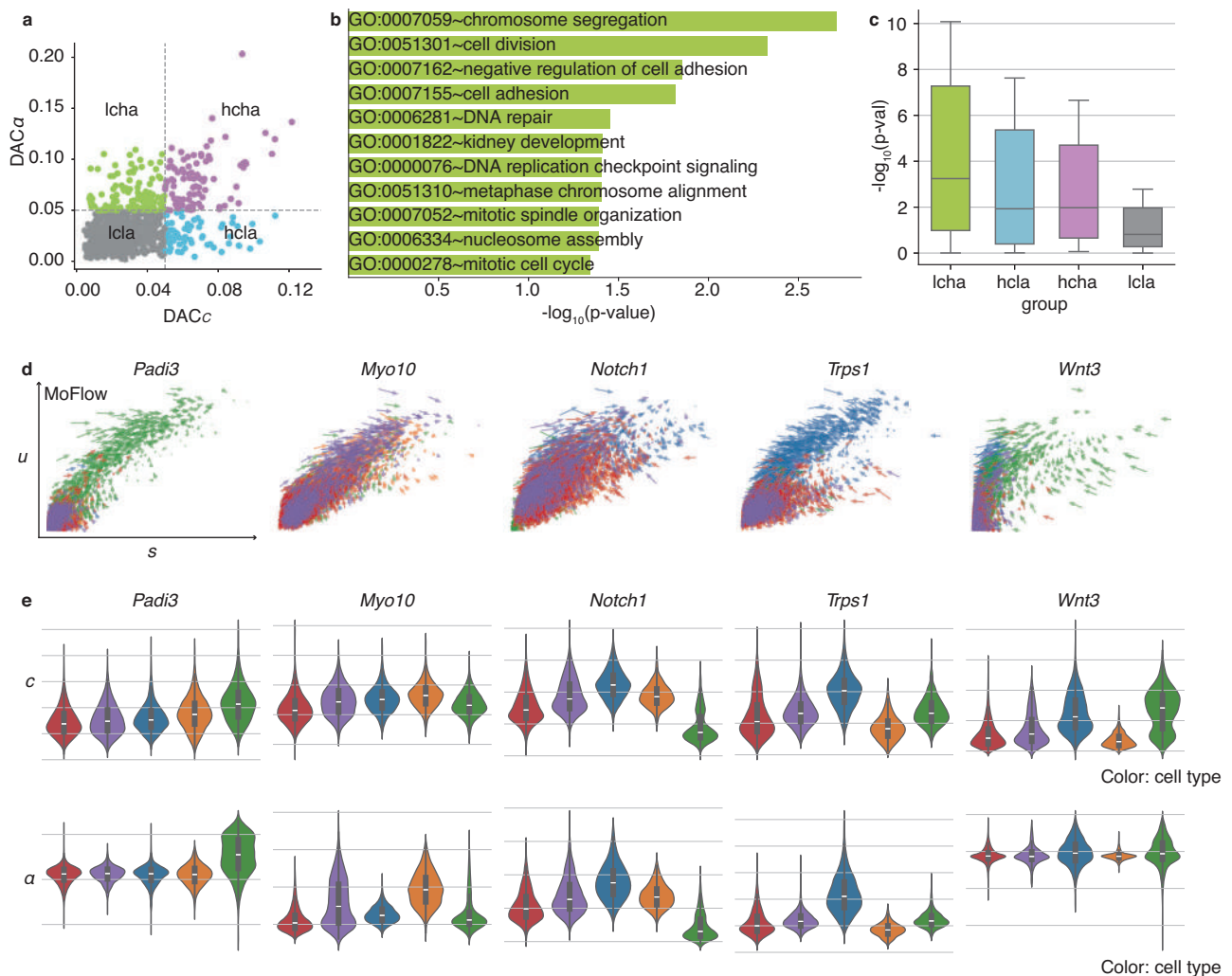


Fig. 5 | Decoupling of transcriptional rate and chromatin accessibility reveals gene-specific regulatory strategies. **a** Genes categorized into four groups based on DAC of chromatin accessibility (c) and transcription rate (a): HCHA, HCLA, LCHA, LCLA. **b** Functional annotation enrichment for the LCHA group using DAVID. LCHA genes are enriched for mitotic regulation. **c** Association of gene groups with cell cycle score. LCHA genes show the strongest link, while LCLA genes show the weakest association with the cell cycle. Two sided t-test was performed for gene

expression between high and low cell cycle score cells. The boxplot boundaries are defined by the first and third quartiles, with the whiskers extending to 0.5 times the interquartile range. The median is marked at the center of the plot. **d** Velocity streamlines for representative genes (*Padi3*, *Myo10*, *Notch1*, *Trps1*, *Wnt3*) predicted using MoFlow. MoFlow captures correct directions in all cases. **e** Cell-type-specific distribution of chromatin accessibility (c) and transcription rate (a) of genes estimated from MoFlow. The median is marked at the center of the plot.

contains cells in an incomplete differentiation state, as indicated by the continued expression of ExM marker genes^{38,39} across both UL and DL populations (Supplementary Fig. 4a).

MoFlow successfully recovered known trajectories, as inferred from velocity vectors and pseudotime estimation (Fig. 6a). Compared to cellDancer, which struggled to accurately reconstruct the differentiation process from IPCs to ExM and subsequently to upper-layer neurons, MoFlow effectively captured these transitions. Moreover, while scVelo, MultiVelo and cellDancer exhibited a biologically implausible backflow pattern, suggesting differentiation from IPCs to RG or Cyc. Prog., MoFlow captured the expected forward transition and downstream lineages with greater fidelity. Pseudotime inferred from the velocity embedding was consistent with the expected progression of Cyc. Prog., as validated by the alignment between pseudotime values and cell cycle scores (Fig. 6b, c)^{7,14}. In particular, cells at early pseudotime displayed elevated G2/M scores, reflecting active proliferation, whereas cells at later pseudotime showed a progressive decline in cycling activity. This correspondence supports the biological plausibility of the inferred trajectories. MoFlow also improved transition coherence within clusters (Fig. 6d)⁴⁰, reducing velocity

discontinuities and producing a biologically smoother and more consistent developmental landscape.

We next compared MoFlow's gene-wise results with MultiVelo's model-based classification. Model 1 genes exhibited significantly higher m_1 scores, while model 2 genes showed elevated m_2 scores, confirming consistency between the two frameworks (Supplementary Fig. 4b). Moreover, MoFlow's continuous RNA-on/off metrics aligned with MultiVelo's discrete state assignments with RNA-off scores being significantly higher in model 2 genes.

To investigate the dynamic RNA regulation, we classified genes into 12 clusters based on their transcriptional changes along pseudotime (Fig. 6e). Functional annotation analysis showed that these genes are enriched in the neuronal development and neural and synaptic function (Fig. 6f). Genes in clusters 0–4 were associated with cell cycle-related processes, particularly cluster 0, which was enriched for cell division, DNA repair, and mitosis, indicating early cell cycle activity. Clusters 4–7 reflected neurodevelopmental processes, such as axon development and cell migration. Clusters 5–9 and 11 were enriched for synaptic functions, including transmission, organization, memory, and sensory processing. In contrast, cluster

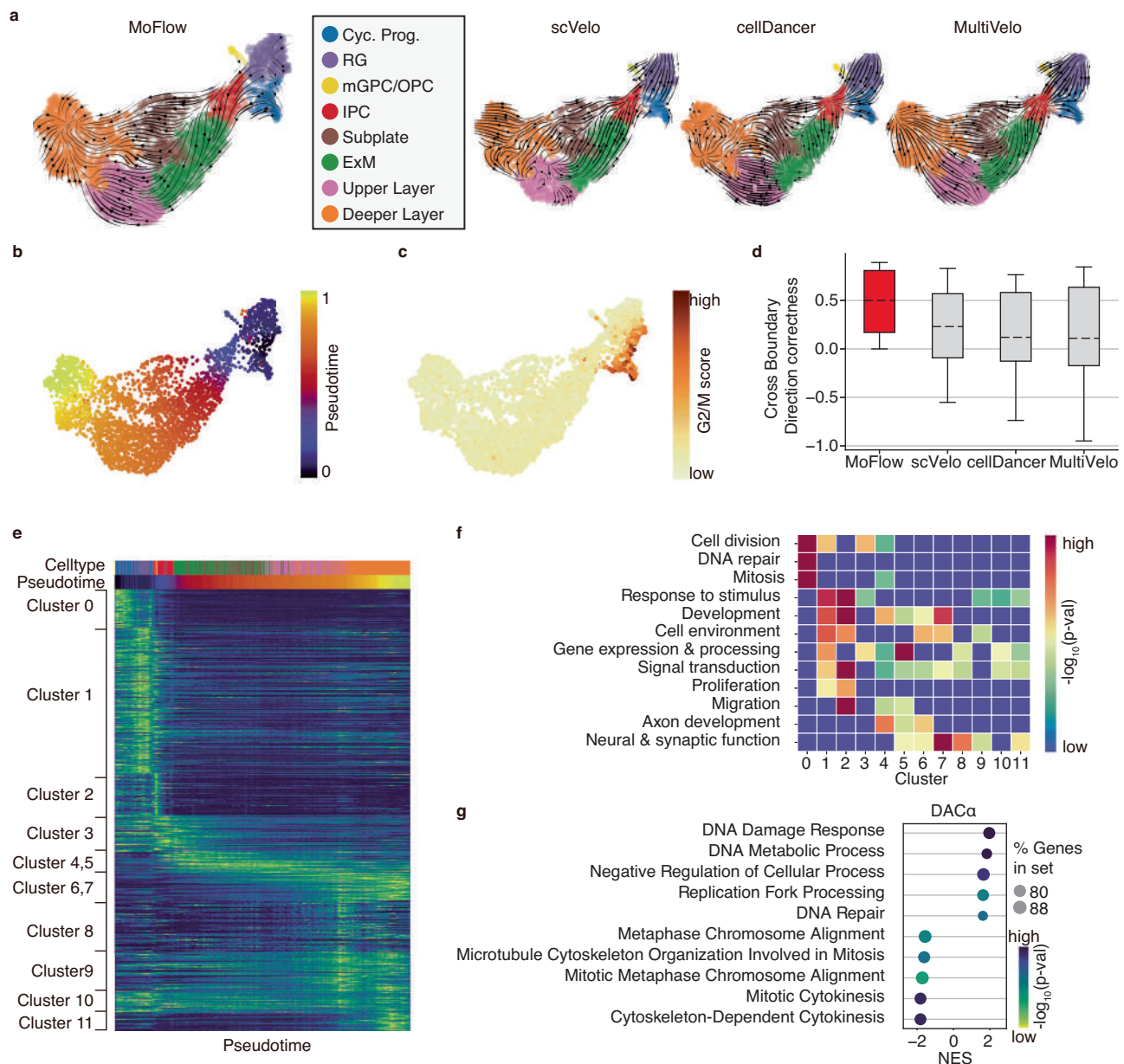


Fig. 6 | MoFlow reveals transcriptional dynamics and regulatory heterogeneity in developing mouse brain. **a** RNA velocity streamlines inferred by MoFlow, scVelo, cellDancer, and MultiVelo on the E18 mouse brain dataset. MoFlow captures expected differentiation trajectories from cycling progenitors (Cyc. Prog.; blue) through RG (radial ganglia; purple) and IPC (intermediate progenitor cell; red) to upper and deep-layer (UL/DL; pink/orange) via ExM (excitatory migrating neuron; green) or subplate (brown). **b** Pseudotime estimated by MoFlow reflects the progression of neurodevelopment. **c** Cell cycle score projected onto the embedding. Cyc. Prog. Cells exhibit the highest cell cycle scores, with elevated G2/M activity that aligns with early pseudotime regions. **d** Cross-boundary direction correctness

(CBDir) comparison across models. MoFlow achieves higher directionality coherence across lineages. The boundaries are defined by the first and third quartiles, with the whiskers extending to 1.5 times the interquartile range. The mean is marked at the center of the plot. **e** Pseudotime-aligned gene expression clusters reveal distinct temporal transcriptional programs. **f** Functional annotation enrichment across clusters using DAVID highlights progression from cell cycle and DNA repair to synaptic and neurodevelopmental functions. **g** GSEA using DAC scores for transcription rate (α) from cluster 0 genes shows enrichment of cell cycle-related pathways, particularly DNA damage response or mitotic regulation, in early progenitor states.

10 showed minimal enrichment for neuronal or developmental pathways, suggesting potential involvement in unrelated or universal regulatory processes.

We then computed DAC scores for both transcription rate (α) or chromatin accessibility (c), then performed GSEA³² on each DAC metric. Genes with negative normalized enrichment scores (NES) in both DAC $_{\alpha}$ and DAC $_{c}$ were predominantly linked to mitotic processes, particularly enriched in Cyc. Prog. cells (Supplementary Fig. 4c, d). Notably, for genes overexpressed in Cyc. Prog. (cluster 0), DAC $_{\alpha}$ showed positive enrichment for DNA repair and metabolism, while

mitotic terms were still negatively enriched (Fig. 6g). In contrast, DAC $_{c}$ enrichment was weaker and less specific, suggesting that transcriptional bursts, rather than chromatin remodeling, may drive acute responses to developmental stress (Supplementary Fig. 4e).

For further validation, we examined genes involved in DNA damage response. These genes showed a clear directional trend from Cyc. Prog. to RG in MoFlow, but not in MultiVelo, where the direction was reversed (Supplementary Fig. 4f, g). This misassignment likely contributed to MultiVelo's erroneous backflow pattern (Fig. 6a). Interestingly, several DNA damage response genes displayed elevated

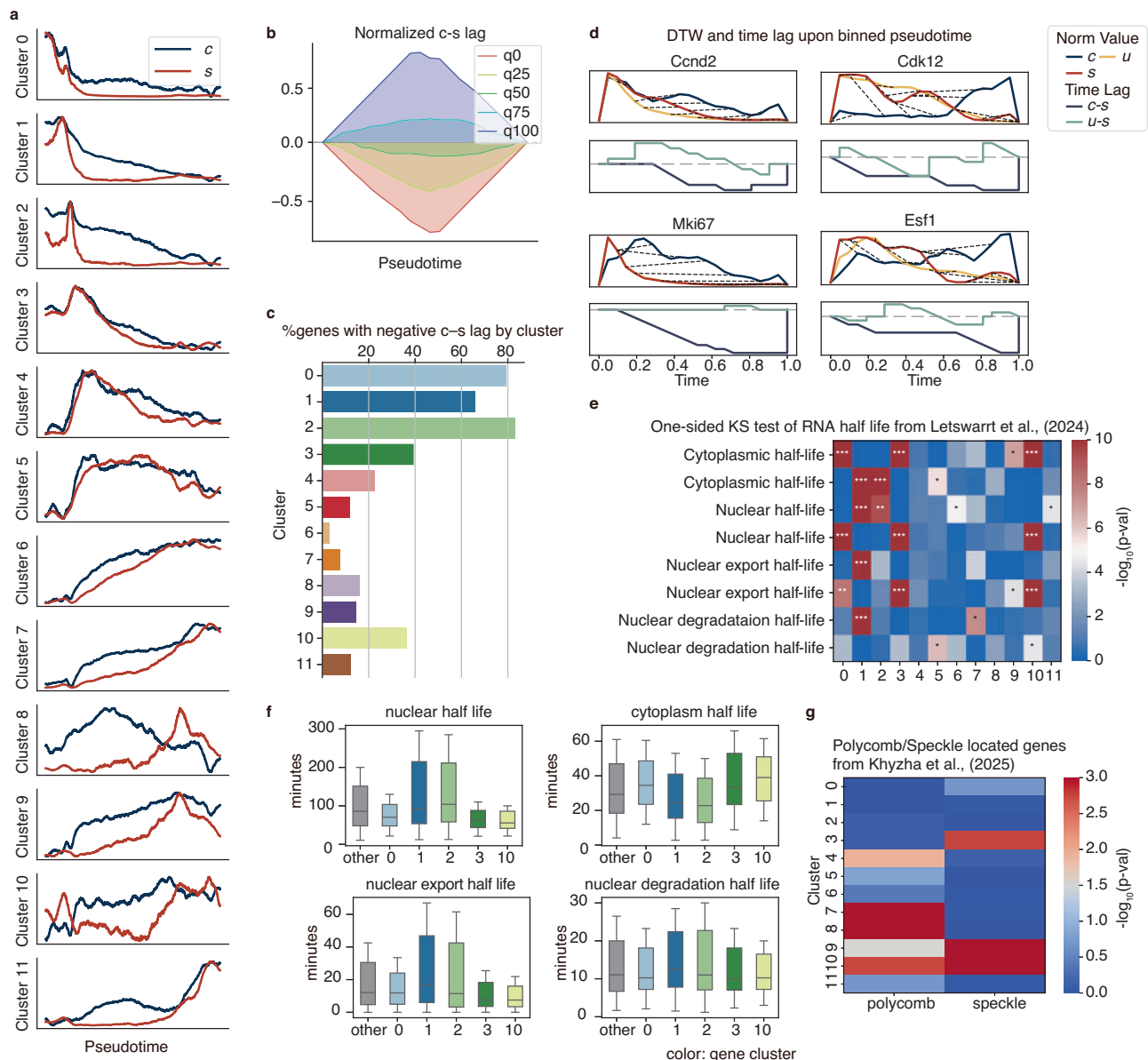


Fig. 7 | Time lag between chromatin accessibility and spliced RNA reveals regulatory heterogeneity driven by RNA kinetics. a Mean chromatin accessibility (*c*) and spliced RNA (*s*) levels for each of 12 gene clusters across pseudotime. **b** Quantiles (*q*) of *c-s* time lags inferred by DTW across pseudotime. The median time lag is negative at most times. **c** Proportion of genes with negative *c-s* lag by cluster. **d** Example genes from lag-enriched clusters show asynchronous *c-s* trajectories. In clusters 0–3 (e.g., *Ccnd2*, *Mki67*), *s* decays ahead of chromatin closure. In cluster 10 (e.g., *Cdk12*, *Esf1*), *s* precedes chromatin opening. **e** One-sided KS test comparing RNA half-life distributions between cluster-specific genes and background. Clusters 0, 3, and 10 show short nuclear half-life and fast nuclear export,

while clusters 1 and 2 show prolonged nuclear retention. **f** Distribution of nuclear, cytoplasmic, export, and degradation half-lives for cluster 0, 1, 2, 3, and 10 compared to other clusters. The boundaries are defined by the first and third quartiles, with the whiskers extending to 0.5 times the interquartile range. The median is marked at the center of the plot. **g** Enrichment of polycomb- and speckle-associated genes by cluster (Fisher's exact test). Cluster 10 is significantly associated with nuclear compartments implicated in transcriptional sequestration and stimulus-responsive export. *** denotes $p < 0.001$, ** denotes $p < 0.005$, * denotes $p < 0.05$.

α levels across RG and subsequent cell types, suggesting that transcriptional activation of repair programs may accompany or immediately follow lineage commitment (Supplementary Fig. 4h). These observations are in line with proposed checkpoint functions of RG-transient activation of stress response programs prior to lineage commitment⁴¹—and with reports that neural stem and progenitor cells, including RG, mount a robust DNA damage response as a protective mechanism during development⁴². Together, these results demonstrate MoFlow's capacity to recover transcriptional programs and cell-type-specific regulatory dynamics with higher fidelity than existing RNA velocity models.

Time lag between chromatin accessibility and spliced RNA suggests regulatory delays shaped by RNA kinetics

To investigate the temporal coordination between chromatin accessibility and gene expression, we examined the average *c* and *s* profiles of clustered genes across pseudotime (Fig. 7a). While most clusters displayed synchronous dynamics, cluster 10 showed a distinct pattern in which *s* accumulation preceded any detectable increase in *c*.

To quantify this phenomenon, we applied DTW between *c* and *s* for each gene and calculated the *c-s* lag (Fig. 7b)^{25,26}. While substantial number of genes exhibited negative *c-s* lags, clusters 0–3 and 10 showed a significantly higher proportion of genes with negative lags

(Fig. 7c). Despite cluster 10 genes showing visible asynchronous (Fig. 7a), clusters 0–3 also demonstrated notable enrichment, suggesting distinct but related mechanisms of temporal asynchrony.

In clusters 0–3, *s* decreased faster than *c*, consistent with a model 2-type transcriptional shutdown in which RNA is degraded or exported before chromatin closure¹². Representative genes such as *Ccnd2* and *Mki67* displayed this pattern, with *s* levels falling sharply while *c* declined more gradually (Fig. 7d).

By contrast, cluster 10 exhibited early *s* accumulation before any rise in *c*. Genes such as *Cdk12* and *Esf1* showed prominent *s* expression ahead of chromatin accessibility (Fig. 7d), suggesting a regulatory program in which transcripts are maintained or rapidly without relying on chromatin remodeling.

Two technical considerations are critical to interpreting these observations. First, as the 10x multiome platform captures nuclear RNA rather than total RNA, apparent degradation may reflect nuclear export. Second, both *c* and *s* were min-max normalized to the [0, 1] range for modeling compatibility, which may obscure low-level chromatin accessibility variance.

To further investigate the basis of negative *c*–*s* lagging, we sought to determine whether gene-specific RNA half-life in the nuclear or cytoplasm could account for this heterogeneity. Therefore, we integrated external measurements of RNA half-lives in each organelle from NIH3T3 cells (Fig. 7e, f)⁴³. Genes from lag-enriched clusters exhibited two distinct kinetic signatures: (i) clusters 0, 3, and 10 showed short nuclear half-life, long cytoplasmic half-life, and fast export; (ii) clusters 1 and 2 showed long nuclear half-life, short cytoplasmic half-life, and slow export. Although our gene clusters were defined in the context of brain development and the half-life measurements were obtained from NIH3T3 cells, the consistent half-life patterns in clusters with high negative time-lag proportions represent a meaningful validation of these kinetic signatures.

Through half-life analysis, we propose that negative *c*–*s* lags in clusters 0–3 arise from two scenarios: (i) delayed chromatin inactivation (e.g., clusters 1, 2), and (ii) rapid RNA turnover or export (e.g., clusters 0, 3). However, cluster 10 still presents unique case: while *c* and *s* remain detectable across pseudotime, *s* rises ahead of *c*, implying that these genes are constitutively expressed and rapidly upregulated in response to cellular cues.

Functional annotation of cluster 10 revealed enrichment for genes involved in transcription, translation, and stimulus response, suggesting that they may function as inducible effectors rather than lineage-specific markers (Fig. 6f). To determine whether time-lagged genes and those various RNA half-life are associated with RNA localization in known nuclear compartment, we compiled gene sets previously known as polycomb or nuclear speckle RNAs⁴⁴ (Fig. 7g). Interestingly, cluster 10 genes shows statistical significance in Fisher's exact test, meaning highly overlapped genes with nuclear compartment RNAs. Although such RNAs are typically nuclear-retained, our data show that cluster 10 genes are characterized by short nuclear half-lives and rapid export—an apparent contradiction.

This discrepancy can be explained in part by conditional export: RNAs may remain sequestered in nuclear bodies under basal conditions and be released upon stimulation, such as development, generating early *s* signals. Additionally, half-life estimation excluded noncoding RNAs, such as *MALAT1* and *XIST*, which are abundant in these compartments. Notably, both *MALAT1* and *XIST* were also assigned to cluster 10. Lastly, nuclear localization and kinetic measurements derived from in vitro cell lines may not fully reflect the in vivo dynamics of developing brain tissue.

Our findings offer a mechanistic explanation for previously observed time lags between transcription factor motif accessibility and downstream gene activation¹². While prior studies noted the existence of such lags, the underlying causes remained unclear. Our study suggests that differences in RNA nuclear export rates, nuclear degradation, and half-life heterogeneity contribute to the emergence of these regulatory delays.

MoFlow reconstructs hematopoietic differentiation trajectories

Lastly, we applied MoFlow to 10x Multiome data of human hematopoietic stem and progenitor cells, which encompass multiple lineages of blood differentiation¹². By integrating chromatin accessibility with gene expression profiles, MoFlow inferred velocity streams that broadly recapitulated the expected hematopoietic hierarchy (Supplementary Fig. 5a). Distinct branches corresponding to erythroid (purple), myeloid (GMP/ProGDC; brown/turquoise), lymphoid (LMPP; green), and megakaryocyte/platelet (MEP/Platelet; green/light yellow) fates emerged from the HSC/MPP (orange/red) populations, in line with canonical lineage bifurcations.

When compared with existing approaches, MoFlow produced velocity fields that were overall better aligned with known differentiation directions. In particular, scVelo and cellDancer exhibited spurious trajectories (Supplementary Fig. 5b). For example, velocity vectors did not consistently originate from HSCs but instead displayed backflow from downstream progenitors, such as megakaryocyte progenitors (ProGDK), into the HSC compartment. By contrast, MultiVelo tended to impose overly linear flows, which obscured natural branching patterns. MoFlow recovered both continuity and divergence of trajectories, although its reconstructions were not always fully consistent across all branches.

We note, however, that MoFlow is not entirely free of limitations. Similar to other velocity-based methods, the vector visualization depends on embedding velocity vectors into a fixed low-dimensional space (e.g., UMAP). This reliance can introduce distortions originating from the embedding rather than the underlying dynamics, and thus, the inferred velocity fields should be interpreted with caution.

To provide a quantitative benchmark, we evaluated CDBir across annotated lineage boundaries (Supplementary Table 1 and Supplementary Fig. 5c). MoFlow attained the highest CDBir scores among the tested methods, outperforming scVelo, cellDancer, and MultiVelo. These results suggest that, despite the caveats, MoFlow captures biologically meaningful transition directions in the human hematopoietic system.

Discussion

Single-cell multi-omic profiling has enabled unprecedented resolution of regulatory dynamics underlying cell fate decisions. However, existing RNA velocity frameworks remain limited in their capacity to disentangle the complex interplay between chromatin accessibility and gene expression, particularly in branching and asynchronous developmental systems. Here, we present MoFlow, a locally adaptive, deep-learning-based velocity model that integrates chromatin accessibility and RNA abundance to infer cell-specific transcriptional kinetics without relying on latent time estimation or fixed gene classifications.

MoFlow outperforms existing models, such as scVelo, MultiVelo and cellDancer in reconstructing developmental trajectories across diverse tissues, including the brain, skin and hematopoietic cells. By aligning local velocity vectors based on nearest-neighbor transitions, it accurately recapitulates known lineage transitions while avoiding biologically implausible reversals often inferred by latent time-based methods. Also, MoFlow captures asynchronous and cell-type-specific gene regulation with high fidelity by estimating kinetic parameters per cell.

Beyond trajectory accuracy, MoFlow revealed a spectrum of transcriptional regulation strategies. Despite MoFlow is inspired by the relay velocity model and its implementation in cellDancer¹¹, it provides comprehensive perspectives by combining chromatin accessibility. For instance, its ability to decouple transcription rate (α) from chromatin accessibility (*c*) of mouse skin differentiation uncovered regulatory states not captured by existing methods. Genes in the LCHA group exhibited elevated α despite stable chromatin, indicating chromatin-independent transcriptional flexibility. Conversely, LCLA genes showed limited activity at both levels, aligning with transcriptionally and epigenetically inert states. These distinctions allowed

MoFlow to capture lineage-specific transcriptional boosts—such as *Padi3* and *Myo10*—that were misclassified by previous models.

We also uncovered transcriptional decoupling in the temporal domain. There was a consistent subset of genes exhibiting negative c - s lags via both global time from MultiVelo and MoFlow. Those genes are mainly enriched in two major groups: (i) genes with early RNA decay relative to chromatin closure, and (ii) genes with early RNA induction prior to chromatin opening. This is supported by nuclear half-life analyses, indicating that the former pattern represents asynchronous repression driven by rapid nuclear export or delayed chromatin shutdown. Second case suggests a stimulus-responsive mechanism involving nuclear export from nuclear compartments. We observed enrichment of short nuclear half-lives and rapid export kinetics in those genes, as well as localization signatures to subnuclear domains known for regulatory sequestration.

Together, these findings demonstrate MoFlow's ability to capture regulatory mechanisms at single-cell resolution—beyond the capabilities of existing approaches. By integrating chromatin context into cell-specific kinetic estimation, MoFlow enables interpretable reconstruction of transcriptional logic across diverse developmental systems, without relying on fixed assumptions about latent time.

Despite these advances, MoFlow has limitations. It does not explicitly model long-range enhancer-promoter interactions, nor does it account for transcriptional memory or motif-level regulation. Additionally, inference remains gene-wise, potentially limiting the detection of coordinated pathway-level programs. Future extensions could incorporate spatial context or protein-level modalities to enrich regulatory resolution.

Overall, MoFlow expands the capacity of velocity modeling to reveal the dynamic and modular nature of gene regulation, positioning it as a powerful tool for decoding lineage specification and transcriptional control in complex single-cell systems.

Methods

Previous approaches: RNA velocity incorporating chromatin accessibility

The original RNA velocity model proposed a system of ODEs to describe the RNA life cycle—from transcription to splicing and degradation—as follows:

$$\begin{aligned}\frac{du_i(t)}{dt} &= \alpha_i(t) - \beta_i(t)u_i(t) \\ \frac{ds_i(t)}{dt} &= \beta_i(t)u_i(t) - \gamma_i(t)s_i(t),\end{aligned}\quad (4)$$

where $u_i(t)$ and $s_i(t)$ denote the concentrations of unspliced (pre-mRNA) and spliced (mature mRNA) transcripts of gene i at time t . The parameters $\alpha_i(t)$, $\beta_i(t)$, and $\gamma_i(t)$ are the gene-specific, time-dependent rates of transcription, splicing, and degradation, respectively. For notational simplicity, we omit the gene index i in the equations that follow unless otherwise specified.

A common assumption in previous RNA velocity models is that α is treated as a constant (as in the velocity model) or as a binary value (as in the scVelo model), while β and γ are assumed to be identical across all cells. La Manno et al.⁶ first introduced this ODE-based RNA velocity model, which was later extended by Bergen et al.⁷ into a dynamical model with the following simplified form^{6,7}:

$$\frac{du(t)}{dt} = \alpha \cdot I\{t < t_{\text{off}}\} - \beta \cdot u(t) \quad (5)$$

$$\frac{ds(t)}{dt} = \beta \cdot u(t) - \gamma \cdot s(t) \quad (6)$$

where $I\{\cdot\}$ is an indicator function. This model divides the gene expression process into two distinct phases: (1) the active phase, in which transcription is active and unspliced RNA is produced, and (2) the repression phase, starting at t_{off} , during which transcription ceases.

Li et al. proposed a novel method called MultiVelo, which integrates chromatin accessibility c , measurements into the ODE model by assuming that transcription rates are influenced by chromatin state¹². A single chromatin accessibility value is defined for each gene and cell by aggregating accessibility signals from the promoter and linked peaks. Because chromatin can theoretically be fully open or closed at a specific locus, chromatin accessibility is normalized to the range $[0, 1]$.

During the opening phase, c approaches 1 at a rate proportional to its current value, governed by a constant α_{c_o} . Conversely, during the closing phase, c decays toward 0 at a constant rate α_{c_c} . The ODE system for chromatin dynamics is thus defined as:

$$\frac{dc(t)}{dt} = \begin{cases} \alpha_{c_o} \cdot (1 - c(t)) & \text{(chromatin opening)} \\ -\alpha_{c_c} \cdot c(t) & \text{(chromatin closing)} \end{cases} \quad (7)$$

Assuming possibly asymmetric opening/closing rates, we write a single piecewise formulation controlled by k (with rates α_{c_o} and α_{c_c}):

$$\frac{dc(t)}{dt} = k \cdot \alpha_{c_o} \cdot (1 - c(t)) - (1 - k) \cdot \alpha_{c_c} \cdot c(t) \quad (8)$$

$$\frac{du(t)}{dt} = \alpha \cdot I\{t_{\text{on}} < t < t_{\text{off}}\} \cdot c(t) - \beta \cdot u(t) \quad (9)$$

$$\frac{ds(t)}{dt} = \beta \cdot u(t) - \gamma \cdot s(t) \quad (10)$$

Here, t_{on} marks the onset of the transcriptional induction phase, which is temporally delayed from the start of chromatin opening.

Previous approaches: relay velocity model

Li et al.¹¹ proposed the relay velocity model and introduced cellDancer, which leverages deep neural networks (DNNs) to estimate kinetic parameters in the RNA velocity model¹¹.

In an scRNA-seq dataset, gene i is observed across n cell snapshots at time points $t = t_1, t_2, \dots, t_n$, where each cell is assumed to represent a different stage of development. For simplicity, the j th cell is denoted as time point t .

For each gene i , the transcription, splicing, and degradation rates are modeled as functions of the unspliced and spliced abundances:

$$\begin{pmatrix} \alpha_i(t) \\ \beta_i(t) \\ \gamma_i(t) \end{pmatrix} = f_i(u_i(t), s_i(t)) \quad (11)$$

In cellDancer, the unknown function f_i is parameterized by a gene-specific deep neural network (DNN) with parameters θ_i :

$$\begin{pmatrix} \alpha_i(t) \\ \beta_i(t) \\ \gamma_i(t) \end{pmatrix} = \Phi_{\theta_i}(u_i(t), s_i(t)), \quad (12)$$

where θ_i denotes the learnable weights of the network.

For simplicity, we omit the gene index i in the equations that follow unless otherwise specified. The dynamics of unspliced and spliced mRNAs are formulated as follows:

$$\frac{u(t + \Delta t) - u(t)}{\Delta t} = \alpha(t) - \beta(t) \cdot u(t) \quad (13)$$

$$\frac{s(t + \Delta t) - s(t)}{\Delta t} = \beta(t) \cdot u(t) - \gamma(t) \cdot s(t) \quad (14)$$

Here, Δt denotes a small time increment, and the transcription (α), splicing (β), and degradation (γ) rates are modeled as functions of

current unspliced and spliced RNA levels for each cell j . A key assumption of the relay velocity model is that neighboring cells in the expression space can be regarded as proxies for the future states of cell j after a short time increment Δt . This “relay” view allows the model to evaluate how well the predicted velocity of a cell aligns with the observed progression toward its neighbors, thus enabling velocity learning without explicit time-series measurements.

The estimated rates are used to extrapolate the RNA states at the next time step. To evaluate how well the predicted RNA velocity vector $\mathbf{v}_j$ aligns with observed transcriptomic progression, cellDancer defines a cosine similarity loss. The progression is approximated by the direction from cell j to one of its neighboring cells j' in expression space:

$$\mathcal{L}_j = \min_{j' \in \mathcal{N}_j} (1 - \cos(\mathbf{v}_j, \mathbf{x}_{j'} - \mathbf{x}_j)), \quad (15)$$

where $\mathcal{N}_j$ denotes the neighborhood of cell j , and $\mathbf{x}_{j'} - \mathbf{x}_j$ represents the transcriptomic displacement toward a neighboring cell. The neighborhood $\mathcal{N}_j$ of each cell j can be defined in a gene-specific phase space. Lastly, total loss for gene i is then defined as:

$$\mathcal{L}_i = \frac{1}{N} \sum_{j=1}^N \mathcal{L}_{ij}, \quad (16)$$

where N denotes the total number of cells to estimate. cellDancer obtains cell-specific parameters θ_i for each gene i by minimizing $\mathcal{L}_i$ using the Adam optimizer.

Extension of relay velocity model to multi-omics: chromatin-aware velocity

Inspired by the relay velocity model and its implementation in cellDancer¹¹, we extend the framework to incorporate chromatin accessibility data alongside unspliced and spliced RNA.

Unlike previous approaches, which estimates two fixed gene-specific constants α_c and α_s to represent chromatin opening and closing rates, our framework leverages cell-specific modeling of chromatin dynamics. Because MoFlow directly learns parameters as a function of both chromatin and RNA states at the single-cell level, it is not necessary to assign separate constants for the opening and closing phases. We therefore introduce a unified rate parameter α_c that governs chromatin accessibility dynamics across both states. The dynamics of chromatin accessibility $c(t)$ is modeled using the following ODE:

$$\frac{c(t + \Delta t) - c(t)}{\Delta t} = \alpha_c(c(t), u(t), s(t)) \cdot (k(t) - c(t)) \quad (17)$$

where $k(t) = 1$ during the chromatin opening phase and $k(t) = 0$ during the closing phase. This formulation, consistent with the chromatin dynamics in MultiVelo, allows the system to smoothly model the transition of chromatin accessibility toward either the fully open or fully closed state, driven by the learned rate parameter $\alpha_c(c(t), u(t), s(t))$.

Similar to previous approaches, MoFlow predicts kinetic parameters by training DNNs that take chromatin accessibility c , unspliced RNA u , and spliced RNA s as inputs. Specifically, MoFlow employs two independent neural networks:

$$\begin{pmatrix} \alpha_c(t) \\ k(t) \end{pmatrix} = \Phi_{\theta_c}(c(t), u(t), s(t)) \quad (18)$$

$$\begin{pmatrix} \alpha(t) \\ \beta(t) \\ \gamma(t) \end{pmatrix} = \Phi_{\theta_{us}}(c(t), u(t), s(t)) \quad (19)$$

Here, Φ_{θ_c} is trained to predict the chromatin opening/closing rate $\alpha_c(t)$ with binary parameter signifying chromatin phase $k(t)$, while $\Phi_{\theta_{us}}$ predicts transcription, splicing, and degradation rates.

Unlike gene expression dynamics, chromatin accessibility evolves slowly, and data–ATACseq–are noisy because of its indirectness, making it challenging to learn the correct chromatin state transition (open or close) directly from data. Therefore, instead of predefining whether chromatin is opening or closing for a given cell, MoFlow infers this state by evaluating which assumption better explains the observed dynamics.

At each training step, we first find an optimal future cell j^* among neighborhood of cell j :

$$j^* = \operatorname{argmin}_{j' \in \mathcal{N}_j^{\text{cus}}} (\mathcal{L}_{j'}^{\text{us}}) \quad (20)$$

$$\mathcal{L}_{j'}^{\text{us}} = (1 - \cos(\mathbf{v}_j^{\text{us}}, \mathbf{x}_{j'}^{\text{us}} - \mathbf{x}_j^{\text{us}})), \quad (21)$$

where $\mathbf{v}_j^{\text{us}}$ denotes the predicted velocity vector for cell j based on unspliced (u) and spliced (s) RNA dynamics, $\mathbf{x}_{j'}^{\text{us}} - \mathbf{x}_j^{\text{us}}$ represents the transcriptomic displacement from cell j toward its neighbor j' in the (c, u, s) expression space. $\mathcal{N}_j^{\text{cus}}$ denotes the neighborhood of cell j defined in the joint chromatin-unspliced-spliced (c, u, s) space. The neighborhood is constructed as the set of closest cells under the Mahalanobis distance using both chromatin accessibility and RNA abundances (default `n_neighbors = 40`). Based on the cell $j^* \in \mathcal{N}_j^{\text{cus}}$ that minimizes this cosine loss, we then evaluate two scenarios for chromatin dynamics–open ($k = 1$) and close ($k = 0$)–and select the one with lower cosine loss against the same neighbor:

$$\mathcal{L}_j^{\text{cus}, k=1} = 1 - \cos(\mathbf{v}_j^{\text{cus}}(k=1), \mathbf{x}_{j^*}^{\text{cus}} - \mathbf{x}_j^{\text{cus}}) \quad (22)$$

$$\mathcal{L}_j^{\text{cus}, k=0} = 1 - \cos(\mathbf{v}_j^{\text{cus}}(k=0), \mathbf{x}_{j^*}^{\text{cus}} - \mathbf{x}_j^{\text{cus}}) \quad (23)$$

where $\mathbf{v}_j^{\text{cus}}$ denotes the predicted velocity vector for cell j based on chromatin accessibility (c), unspliced (u) and spliced (s) RNA dynamics, $\mathbf{x}_{j^*}^{\text{cus}} - \mathbf{x}_j^{\text{cus}}$ represents the displacement from cell j toward its optimized neighbor j^* .

In the initial training phase (default: `n_epochs = 20`), the chromatin-related loss is computed as the average of both open and close scenarios:

$$\mathcal{L}_j^{\text{total}} = \frac{1}{2} (\mathcal{L}_j^{\text{cus}, k=1} + \mathcal{L}_j^{\text{cus}, k=0}) \quad (24)$$

After this warm-up period, only the loss corresponding to the better-fitting chromatin state (opening or closing) is retained:

$$\mathcal{L}_j^{\text{total}} = \min(\mathcal{L}_j^{\text{cus}, k=1}, \mathcal{L}_j^{\text{cus}, k=0}) \quad (25)$$

This dynamic formulation enables MoFlow to flexibly infer context-specific chromatin states and gene regulatory kinetics without requiring explicit supervision or prior annotations, making it well-suited for modeling complex multi-omic cell state transitions. Total loss for gene i is then defined as:

$$\mathcal{L}_i = \frac{1}{N} \sum_{j=1}^N \mathcal{L}_{ij}^{\text{total}}, \quad (26)$$

where N denotes the total number of cell to estimate. To estimate kinetic parameters for given gene i , MoFlow minimizes the total loss $\mathcal{L}_i$ for each gene using the Adam optimizer. The architecture of the DNNs is as follows: an input layer with $3N$ nodes; three fully connected hidden layers with 64, 48, and 32 nodes respectively, each

using the ReLU activation function; and an output layer with either N nodes (for Φ_{θ_c}) or $3N$ nodes (for $\Phi_{\theta_{us}}$). A sigmoid activation is applied at the output layer to constrain the predicted parameters within the range $[0, 1]$. The learning rate is set to 0.001, and the weight decay is 0.04 to apply L2 regularization and prevent overfitting. Training is monitored with an early stopping mechanism that checks the loss every 10 epochs with a maximum of 200 epochs. All models in the paper are trained with this default configuration, though users can modify these hyperparameters via the command line interface. MoFlow is implemented in PyTorch Lightning, consistent with the implementation of cellDancer.

Data preprocessing

All data in this study were downloaded publicly (see details in the “Data availability” section).

Human brain dataset. We utilized the preprocessed dataset provided by Li et al.¹², which includes chromatin signal smoothing via the Weighted Nearest Neighbor (WNN) approach to integrate multimodal information⁴⁵. Instead of redefining clusters as done in the original study, we retained the cluster annotations as designated in the source publication.

SHARE-seq mouse skin dataset. We utilized the preprocessed dataset provided by Li et al., which includes chromatin signal smoothing via the WNN approach to integrate multimodal information.

10x embryonic mouse brain dataset. We followed a preprocessing strategy similar to that of Li et al., using a combination of RNA and ATAC-seq data from the 10x Genomics multiome platform. Briefly, RNA reads were processed with velocity to quantify unspliced and spliced counts, followed by filtering, normalization, and neighbor graph construction using scanpy and scVelo. Chromatin accessibility was quantified by aggregating distal enhancer and promoter peaks correlated with gene expression, and normalized using the TF-IDF method⁴⁶. To account for the sparsity of ATAC-seq data, we applied the WNN approach from Seurat v4 to smooth chromatin signals across multimodal neighbors. These processed RNA and chromatin matrices were then used for downstream dynamical modeling.

Human HSPC dataset. We utilized the preprocessed dataset provided by Li et al., which includes chromatin signal smoothing via the WNN approach to integrate multimodal information¹¹.

Data analysis

Post-fitting analyses. Prior to downstream analyses, scVelo dynamical modeling, cellDancer and MultiVelo were applied to the same pre-processed inputs utilized by MoFlow, with all default parameters retained. Specifically, for scVelo and cellDancer, only the transcriptomic component of the MoFlow input was provided, omitting ATAC-derived chromatin information; in contrast, MultiVelo received the complete multi-omic dataset (both RNA and ATAC) in exactly the same format as used by MoFlow.

To investigate cellular dynamics within our multi-omic velocity framework, we employed downstream tools from the scVelo suite. Cell-to-cell transition probabilities were computed via `tl.velocity_graph` applied to each method’s predicted velocity vectors. For MoFlow, `tl.velocity_graph` was invoked with `xkey = “s”` (denoting the normalized spliced count matrix) and `vkey = “velo_s”` (the layer containing MoFlow-derived spliced RNA velocities). In the case of cellDancer, the output—a pandas DataFrame of per-cell spliced RNA velocity estimates—was converted into an AnnData object to ensure compatibility with the scVelo pipeline.

Subsequent to computing cell-to-cell transition probabilities, cellular pseudotime trajectories were inferred using `tl.velocity_pseudotime` for both MoFlow and cellDancer. For MultiVelo, global latent time was obtained via `tl.latent_time`.

Model evaluation metrics. To assess the model performance within trajectories, we used CBDiR proposed previously⁸. CBDiR quantifies the accuracy of predicted transitions from a source cluster to a target cluster by evaluating the alignment of velocity vectors at the cluster boundary. Boundary cells are defined as those within the source cluster that neighbor cells in the target cluster, reflecting short-term developmental trajectories.

Given a set of cells $c' \in C_A \cap N(c)$ in the target cluster C_A and their neighbors $N(c)$, Per-cell CBDiR for c (CBDiR(c)) is defined as:

$$\text{CBDiR}(c) = \frac{1}{\text{Norm}} \sum_{c' \in C_A \cap N(c)} \cos(\mathbf{v}_c, \mathbf{x}'_c - \mathbf{x}_c) \quad (27)$$

$$\text{Norm} = |c' \in C_A \cap N(c)| \quad (28)$$

where:

$\mathbf{v}_c$ is embedded velocity vector of cell c ,

$\mathbf{x}_c$ and $\mathbf{x}'_c$ are the coordinates embedded position of cell c and c' .

Since CBDiR relies on known developmental directions (i.e., ground truth transitions), it serves as a biologically meaningful measure to validate the directionality inferred by RNA velocity models.

Quantification of transcriptional states. To model transcriptional repression dynamics, we quantified gene-wise transcriptional states and discordances between chromatin accessibility and RNA production. For each gene, we computed the following four scores: RNA-on score, RNA-off score, m_1 score and m_2 score.

RNA-on and RNA-off scores. Let v_j^u and v_j^s denote the predicted unspliced and spliced RNA velocity for cell j , respectively. For each gene, we define:

$$\text{RNA-on score} = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \mathbb{I}(v_j^u > 0 \wedge v_j^s > 0) \quad (29)$$

$$\text{RNA-off score} = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \mathbb{I}(v_j^u < 0 \wedge v_j^s < 0) \quad (30)$$

where $\mathcal{J}$ represents the set of cells with non-negligible expression for the given gene, defined as those for which c_j , u_j , and s_j are greater than threshold times the maximum value of the corresponding quantity observed for that gene across all cells (default threshold: 0.1).

m_1 and m_2 scores. To examine potential discordance between chromatin accessibility and transcriptional activity, we focused on cells with high expression of both unspliced and spliced RNA:

$$\mathcal{H} = \left\{ j \mid u_j > P_{90}(u), s_j > P_{90}(s) \right\}, \quad (31)$$

where $P_{90}(u)$ and $P_{90}(s)$ denote the 90th percentiles of unspliced and spliced expression across cells. This high-expression threshold focuses the analysis on the transcriptional peak, where transitions into repression are most likely to occur. Since model 1 and model 2 dynamics are expected to manifest near the initiation of repression, restricting to high unspliced and spliced RNA levels ensures that m_1 and m_2 scores capture these transition points accurately. Notably, we tested the appropriate default threshold for scoring and set it as the 90th percentile (Supplementary Fig. 6a).

Within this high-expression subset $\mathcal{H}$, we define:

$$m_1 = \frac{1}{|\mathcal{H}|} \sum_{j \in \mathcal{H}} \mathbb{I}(v_j^c < 0 \wedge v_j^u > 0) \quad (32)$$

$$m_2 = \frac{1}{|\mathcal{H}|} \sum_{j \in \mathcal{H}} \mathbb{I}(v_j^c > 0 \wedge v_j^u < 0) \quad (33)$$

Here, v_j^c denotes the chromatin accessibility velocity for cell j . The m_1 score reflects cases where unspliced RNA is increasing despite chromatin closing (possible delayed repression), while the m_2 score reflects decreasing RNA despite chromatin opening (possible delayed induction). For visualization and gene classification, m_1 and m_2 are normalized such that $m_1 + m_2 = 1$ for each gene.

Notably, the RNA-on and RNA-off scores are suitable for identifying genes with consistent transcriptional activation or repression across cells, making them appropriate for characterizing gene states or comparing to discrete kinetic models (e.g., MultiVelo's "on/off" states). In contrast, the m_1 and m_2 scores are designed to capture discordance between chromatin accessibility and transcriptional activity during transcription repression initiation. Those are useful to identify genes undergoing potential repression or delayed regulatory transitions and are therefore appropriate for investigating mechanisms such as transcriptional decoupling, asynchronous repression, or chromatin-RNA timing mismatches. For downstream applications, both score types offer complementary insights: RNA-on/off scores reflect dominant transcriptional behavior, while m_1/m_2 scores highlight regulatory conflict or complexity.

Projection of external datasets and annotation transfer. We analyzed an external scRNA-seq dataset derived from the Allen Brain Map (<https://portal.brain-map.org/atlas-and-data/rnaseq>). Pre-processed and normalized gene expression data were read into an AnnData object using Python and the Scanpy framework. To enable comparison and integration with human brain development data, we first computed the top 50 principal components from the Allen Brain Map dataset using PCA. The external RNA data were projected into this PCA space using the same gene loadings, ensuring a shared feature representation. Next, we embedded both datasets into a shared low-dimensional UMAP space. The UMAP model was fitted on the external PCA representation and applied to the development data PCA projections using the pretrained UMAP transformation.

Dynamic time warping (DTW) analysis. To quantify timing discrepancies between chromatin accessibility and RNA expression dynamics, we applied DTW, a nonlinear alignment method that identifies the optimal warping path between two time series²⁵. This approach allows us to detect regulatory asynchrony—such as delays between chromatin opening and RNA induction—without assuming that changes occur simultaneously.

We implemented DTW using the `fastdtw` Python package²⁶. For each gene, we aggregated chromatin accessibility (c), unspliced RNA (u), and spliced RNA (s) profiles into equally sized bins (default: $n_{\text{bins}} = 20$) along pseudotime. When using MoFlow predictions, pseudotime was based on global progression; for MultiVelo, either global or latent time was used depending on the analysis. To improve robustness, pseudotime values were clipped at the 95th percentile to remove outliers.

To reduce noise and ensure comparability, expression profiles were processed as follows:

- Time series were digitized into equally sized bins along pseudotime
- Expression values within each bin were averaged to create binned profiles

- Binned expression profiles were normalized to the range [0, 1] using min-max scaling
- Normalized profiles were smoothed using a uniform filter (default window size = 3 bins)
- Smoothed profiles were padded with zeros at both ends to stabilize DTW

DTW was then applied to pairwise combinations of modalities—specifically, chromatin vs. spliced RNA (c - s) and unspliced vs. spliced RNA (u - s)—using Euclidean distance as the default distance metric. For each matched point pair (i, j) in the warping path, we computed the time lag as the difference in normalized pseudotime between the reference signal (spliced RNA) and the query signal (chromatin or unspliced RNA). A positive lag indicates that the spliced RNA signal follows the chromatin or unspliced RNA signal, while a negative lag indicates the opposite pattern.

We provide the function `get_dtw()` for performing DTW analysis and `plot_dtw()` for visualization, which allows users to display aligned trajectories, warping paths, and lag curves for individual genes.

Distribution of averages across cell types. To quantify the cell type-specific variability of each gene's kinetic or epigenetic parameter (e.g., transcription rate α or chromatin accessibility c), we defined the distribution of average across cell types (DAC) score as follows.

For each gene g and specific parameter x , we first compute the mean value within each cell type:

$$\mu_{x,g,k} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{i,g},$$

where C_k denote the set of cells annotated as cell type k .

We then define the DAC score for gene g as the variance of these cell-type means:

$$\text{DAC}_g^{(x)} = \frac{1}{K} \sum_{k=1}^K (\mu_{x,g,k} - \bar{\mu}_{x,g})^2, \quad \bar{\mu}_{x,g} = \frac{1}{K} \sum_{k=1}^K \mu_{x,g,k},$$

where K denotes the number of cell types. A high DAC score indicates that the parameter x varies substantially across cell types, reflecting cell type-specific regulation of gene g .

Functional annotation and gene set enrichment analysis. Functional annotation analysis was performed using DAVID²⁰, and GSEA was conducted using the GSEAPy package³². For both approaches, enrichment was evaluated exclusively against the Biological Process (GO: BP) ontology of the Gene Ontology resource^{21,22}. The detailed results of these analyses are provided in Supplementary Data 1–3.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Human brain multi-ome¹³ can be found at the GEO (GSE162170) and the authors' Github Page (<https://github.com/GreenleafLab/brainchromatin>). 10x embryonic mouse brain can be accessed on the 10x Genomics website at <https://www.10xgenomics.com/resources/datasets/fresh-embryonic-e-18-mouse-brain-5-k-1-standard-1-0-0>. SHARE-seq mouse skin dataset²⁸ can be found at the GEO (GSE140203). 10x Human HSPC¹² can be found at the GEO (GSE209878).

Code availability

MoFlow is implemented in Python. The code with the Demo notebooks are available on GitHub (<https://github.com/AriHong/MoFlow>) or Zenodo (<https://doi.org/10.5281/zenodo.17666878>)⁴⁷.

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Author contributions

A.H. conceived projects, performed data collection, development and refinement of code, computational analyses, and wrote the article. S.L. and K.K. supervised the work.

Competing interests

The authors declare no competing interests.

Additional information

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